

FORM 2

THE PATENTS ACT, 1970

[39 of 1970]

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THE PATENTS RULES, 2003

COMPLETE SPECIFICATION

[See section 10 and rule 13]

“AI-ENABLED URBAN FLOOD DETECTION AND EARLY WARNING SYSTEM USING CCTV”

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PREAMBLE OF THE DESCRIPTION

The following specification particularly describes the invention and the manner in which it is to be performed.

AI-ENABLED URBAN FLOOD DETECTION AND EARLY WARNING SYSTEM USING CCTV

FIELD OF THE INVENTION

The present invention generally relates to the field of disaster management systems, computer vision, and smart city infrastructure. More particularly, the invention pertains to an artificial intelligence-enabled system for detecting urban flooding conditions and generating early warnings using closed-circuit television (CCTV) surveillance networks. The invention integrates real-time video analytics, environmental sensing, data communication, and predictive modeling to enable timely identification of flood events and proactive response measures.

BACKGROUND OF THE INVENTION

Urban flooding has become a significant concern in many cities due to rapid urbanization, inadequate drainage systems, and climate variability. Flooding in urban areas leads to severe consequences, including damage to infrastructure, disruption of transportation, and threats to human safety. Traditional flood monitoring systems rely on physical sensors such as water level gauges and rainfall measurement devices, which may provide limited spatial coverage and delayed information.

CCTV surveillance networks are widely deployed in urban environments for security and traffic monitoring purposes. These networks offer extensive coverage and continuous monitoring capabilities, presenting an opportunity to leverage existing infrastructure for flood detection. However, conventional use of CCTV systems is limited to manual observation, which is inefficient and prone to delays.

Recent advancements in artificial intelligence, particularly in computer vision, enable automated analysis of video data. Deep learning models can be trained to identify patterns and anomalies in images and videos, making them suitable for detecting flood conditions. However, existing systems often lack integration with real-time warning mechanisms and predictive analytics.

Another limitation is the absence of systems that combine visual data with environmental factors such as rainfall intensity and drainage capacity. Integrating multiple data sources can improve the accuracy and reliability of flood detection. Additionally, there is a need for scalable systems capable of handling large volumes of video data from multiple cameras.

Existing flood warning systems often do not provide localized alerts, limiting their effectiveness in urban settings where conditions may vary significantly across different areas. Furthermore, ensuring timely communication of warnings to relevant authorities and the public is critical for effective response.

Therefore, there is a need for an AI-enabled system that utilizes CCTV networks for real-time flood detection and integrates predictive analytics and communication mechanisms to provide early warnings and support decision-making.

OBJECTS OF THE INVENTION

The primary object of the present invention is to provide an AI-enabled system for detecting urban flooding using CCTV data.

Another object of the invention is to enable real-time monitoring and analysis of video streams. A further object of the invention is to provide early warning alerts based on detected flood conditions. Another object of the invention is to integrate environmental data for improved accuracy. Yet another object of the invention is to support scalable deployment

across urban areas. A further object of the invention is to enhance disaster preparedness and response.

SUMMARY OF THE INVENTION

The present invention discloses an AI-enabled urban flood detection and early warning system comprising a video acquisition module, a preprocessing module, a computer vision-based detection engine, an environmental data integration module, a predictive analytics module, and an alert generation module.

The detection engine analyzes video streams to identify flood conditions, while the integration module combines visual data with environmental parameters. The predictive module forecasts potential flooding, and the alert module generates notifications for authorities and the public. The system enables real-time monitoring and proactive response.

DETAILED DESCRIPTION OF THE INVENTION

The present invention provides a comprehensive and intelligent system for detecting urban flooding conditions and generating early warnings using closed-circuit television (CCTV) networks integrated with artificial intelligence and predictive analytics. The system is designed to leverage existing urban surveillance infrastructure to enable continuous monitoring and rapid response to flood events.

In accordance with an embodiment of the present invention, the system comprises a video acquisition module configured to capture real-time video streams from multiple CCTV cameras deployed across urban areas. The cameras may be positioned at strategic locations such as roads, intersections, drainage points, and low-lying areas prone to flooding. The module supports various video formats and ensures seamless integration with existing surveillance systems.

The captured video data is transmitted to a preprocessing module, which performs operations such as frame extraction, noise reduction, and image enhancement. The preprocessing module ensures that the video frames are suitable for analysis by improving clarity and reducing artifacts. Techniques such as contrast adjustment and filtering may be applied to enhance the visibility of water accumulation.

The processed video frames are then input into a computer vision-based detection engine, which serves as the core analytical component of the system. The detection engine employs deep learning models, such as convolutional neural networks, to analyze visual patterns and identify the presence of water on surfaces. The models are trained to recognize features associated with flooding, including water reflections, ripples, and changes in surface appearance.

In an embodiment, the detection engine utilizes segmentation techniques to differentiate between flooded and non-flooded areas within the video frames. This enables the system to estimate the extent of flooding and monitor changes over time. The engine may also incorporate temporal analysis to track the progression of water levels.

The system further includes an environmental data integration module that collects data from external sources such as weather stations, rainfall sensors, and drainage systems. This module integrates environmental parameters with visual data to enhance the accuracy of flood detection. For example, high rainfall intensity combined with visual indicators of water accumulation may increase the confidence of flood detection.

In another embodiment, the system includes a predictive analytics module that uses historical and real-time data to forecast potential flooding events. The module employs machine learning algorithms to analyze trends and identify patterns that may lead to flooding.

Predictive models may consider factors such as rainfall patterns, drainage capacity, and historical flood data.

The system also includes an alert generation module that provides early warnings based on detected or predicted flood conditions. The alerts may be delivered to authorities, emergency services, and the public through various communication channels, including mobile applications, SMS, and web platforms. The alerts may include information such as location, severity, and recommended actions.

In an embodiment, the system includes a centralized monitoring platform that aggregates data from multiple cameras and sensors. The platform provides a dashboard interface for monitoring flood conditions in real time. The dashboard may include visualizations such as maps, graphs, and alerts to facilitate decision-making.

The system is designed to be scalable, allowing deployment across large urban areas with numerous CCTV cameras. The architecture supports distributed processing and cloud-based storage, enabling efficient handling of large volumes of data.

Data security and privacy are integral components of the system. The invention implements encryption and access control mechanisms to protect video data and ensure compliance with regulations. The system may also anonymize video data to protect individual privacy.

In operation, video streams are continuously analyzed by the detection engine to identify flood conditions. Environmental data is integrated to improve accuracy, and predictive models forecast potential flooding. When a flood condition is detected or predicted, the system generates alerts and provides real-time updates through the monitoring platform.

The present invention provides significant advantages over conventional flood monitoring systems, including real-time detection, improved accuracy through data integration, predictive capabilities, and efficient use of existing infrastructure. These features enable timely response and reduce the impact of urban flooding.

Accordingly, the invention represents a novel and effective solution for urban flood detection and early warning, contributing to improved disaster management and public safety.

Conventional flood monitoring methods used to assess water levels are often slow and inefficient, especially in rapidly changing urban environments. These approaches usually depend on manual observation or limited sensors, which can lead to delayed response and inaccurate flood assessment.

We present an innovative approach for monitoring urban floods using an AI-enabled CCTV-based system. The proposed method uses existing surveillance cameras and analyzes live video feeds through YOLOv8 in Python to detect water accumulation and estimate flood levels in real time. A Raspberry Pi 4 is used as an edge device to process the detection results and host the web dashboard 24/7 for continuous monitoring. In addition, all flood detection details are stored in a XAMPP MySQL database, enabling data logging, historical analysis, and effective disaster management.

Conducting urban flood monitoring has always been challenging, even though it is critical for public safety and disaster management. Traditional methods often fail to provide real-time and accurate data, leading to delayed responses and increased risk. The proposed system introduces an AI-Enabled CCTV-Based Flood Detection System using YOLOv8 in Python to analyze live video feeds and detect water accumulation automatically.

This system represents a significant improvement by utilizing computer vision and machine learning for accurate flood level estimation. A Raspberry Pi 4 is used as an edge device to process data and host a web dashboard 24/7 for continuous monitoring. All flood detection details are stored in a XAMPP MySQL database, enabling real-time logging, historical analysis, and data visualization.

The integration of real-time analysis, automated alerts, and web-based monitoring improves response time and supports better decision-making. With features like dashboards, alerts, and remote access, the system becomes more practical for authorities and disaster management teams. Overall, this solution provides a scalable, efficient, and cost-effective approach to modern urban flood monitoring.

The AI-Enabled CCTV-Based Flood Detection System is used to monitor and detect urban flood conditions in real time using computer vision and machine learning techniques.

1. The system captures live video feeds from CCTV cameras installed in urban areas such as roads and public spaces.
2. The video stream is processed using YOLOv8 in Python to analyze frames and detect water accumulation on roads.
3. The system calculates the flood percentage and classifies the flood level (Normal, Moderate, Severe) based on detected water coverage.
4. A Raspberry Pi 4 is used as an edge device to process data and host the web dashboard continuously in 24/7 mode.
5. All detection results, including camera ID, flood percentage, time, and status, are stored in a XAMPP MySQL database.

6. The system displays real-time flood data, camera feeds, and alerts on a web dashboard for monitoring and decision-making
7. Automated alerts (SMS/Telegram/Web) are generated when flood levels exceed safe thresholds, enabling faster response and improved disaster management.
8. Authorities can view historical data and trends from the database to analyze flood patterns and take preventive actions.

Data Security and Privacy: All flood detection data, including camera details and monitoring information, is securely stored in a XAMPP MySQL database, ensuring protection from unauthorized access.

System Permissions and Access Control: Authorized users such as administrators and authorities can access the system dashboard, ensuring controlled and secure usage of monitoring data.

Video Data Acquisition and Database Setup: The system collects live video feeds from CCTV cameras, and a structured database is created to store flood detection details like camera ID, time, and flood percentage.

AI Model Integration (YOLOv8): The core process involves using YOLOv8 in Python to analyze video frames and detect water accumulation accurately.

Real-Time Flood Detection and Classification: The system processes frames continuously and classifies flood levels into Normal, Moderate, and Severe based on detected water coverage.

Edge Processing with Raspberry Pi 4: A Raspberry Pi 4 is used to run the detection model and host the web application in 24/7 mode, ensuring continuous monitoring.

Real-Time Feedback and Alert System: The system provides instant feedback through a live dashboard and sends alerts (SMS/Telegram/Web) when flood levels exceed thresholds.

Data Monitoring and Analysis: All detection data is stored and analyzed to identify flood patterns, trends, and high-risk areas for better decision-making.

Technology Platform Integration: The system is integrated with web-based platforms, enabling live monitoring, remote access, and data visualization for authorities and users.

Managing large amounts of real-time data from multiple CCTV cameras is a major challenge, as continuous video processing generates a high volume of flood detection records. Without proper management, this can lead to slow system performance, data loss, and difficulty in retrieving useful insights.

XAMPP MySQL database is implemented to efficiently store and organize flood detection data such as camera ID, flood percentage, timestamp, and status. The system uses optimized database design techniques like indexing and structured tables to ensure fast data retrieval. Old or unnecessary data can be periodically cleaned or archived to reduce storage load. Additionally, only essential detection results (not full video) are stored, minimizing memory usage while maintaining useful information for analysis and visualization.

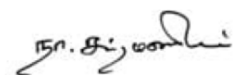
The proposed system can be further enhanced by integrating advanced Artificial Intelligence (AI) models to predict flood conditions before they occur. By analyzing historical flood data stored in the XAMPP MySQL database, along with real-time inputs such as rainfall and water accumulation patterns, the AI system can forecast potential flood risks.

This AI-based approach can dynamically adjust alert levels based on detected trends and environmental conditions. It will help authorities take preventive measures in advance, improve decision-making, and enhance overall disaster management efficiency.

We Claim:

1. An AI-enabled urban flood detection system comprising a video acquisition module, a preprocessing module, a detection engine, an integration module, a predictive module, and an alert module.
2. The system as claimed in claim 1, wherein the video acquisition module captures CCTV data.
3. The system as claimed in claim 1, wherein the detection engine utilizes deep learning models.
4. The system as claimed in claim 1, wherein the system performs segmentation of flooded areas.
5. The system as claimed in claim 1, wherein the system integrates environmental data.
6. The system as claimed in claim 1, wherein the system includes predictive analytics.
7. The system as claimed in claim 1, wherein the alert module provides early warnings.
8. The system as claimed in claim 1, wherein the system includes a monitoring platform.
9. The system as claimed in claim 1, wherein the system supports scalable deployment.
10. The system as claimed in claim 1, wherein the system ensures data security.

Dated this **04th** day of **April 2026**



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AI-ENABLED URBAN FLOOD DETECTION AND EARLY WARNING SYSTEM USING CCTV

ABSTRACT

Urban flooding makes real-time monitoring difficult, causing delayed response and damage. Traditional methods are slow, costly, and often inaccurate. This project proposes an AI-based flood detection system using CCTV cameras and YOLOv8 in Python to detect water levels from live video feeds. A Raspberry Pi 4 is used to host the web dashboard continuously in 24/7 mode for real-time monitoring. The system stores flood detection details in a XAMPP MySQL database for analysis and record maintenance. This cost-effective solution uses existing infrastructure, live alerts, and database support to improve disaster management.